

Topology Determines Emergent Self-Prediction in Multi-Agent Communication Networks: A Scaling Law for Consciousness Correlate Collapse

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Abstract

This paper presents *consim*, a multi-agent simulation that studies how self-prediction emerges from lossy predictive communication on graph topologies. The system is inspired by the Machine Consciousness Hypothesis (Fitz, 2025). It places distributed agents on a network where they communicate through noisy channels and learn to predict their neighbors’ states, but never their own. I measure four consciousness-relevant quantities from the MCH framework: integration (Φ), reflexivity (R), temporal persistence (T), and causal efficacy (E). Across 4,180 simulation runs on five topologies, five noise levels, five grid sizes (12, 18, 24, 36, 48), and three activation functions (tanh, linear, ReLU), I report three main results. First, on high-connectivity grids, causal efficacy anti-correlates with self-prediction ($r = -0.71$) and with temporal persistence ($r = -0.82$). Self-determination destroys self-knowledge, which inverts assumptions in IIT (Tononi, 2004), MCH (Fitz, 2025), and autonomy-based consciousness frameworks. This trade-off holds across all three activations (tanh -0.82 , linear -0.77 , ReLU -0.76) and persists identically in a null model without learning ($r = -0.79$), making it a topological invariant rather than a learning artifact. Second, PCA on the five metrics shows they collapse to a single principal component on Moore grids (82.5% variance explained), forming an “autonomy-predictability axis.” Third, this collapse follows a scaling law: $PC1 = -0.136 + 0.124 \times \ln(N)$ ($R^2 = 0.987$) across five grid sizes (144 to 2,304 agents), with a sharp inflection between $N = 324$ and $N = 576$ agents. Random topologies show no collapse at any size. I present a formal argument that this works like an uncertainty principle for networked systems: on K -regular graphs with bounded activations and correlated neighbors, variance reduction through neighbor-averaging forces a trade-off between causal autonomy and predictability. The constraint on consciousness correlates is an emergent property of structured networks that only appears above a critical system size. Code and data: github.com/brian-mwirigi/consim.

Keywords: consciousness correlates, integrated information, multi-agent systems, emergent self-prediction, network topology, scaling laws, dimensionality collapse

1. Introduction

Theories of consciousness use multiple measurable correlates: integration (Tononi, 2004; Tononi et al., 2016), temporal stability (Baars, 1988), causal efficacy (Seth, 2014; Seth & Bayne, 2022), and self-modeling (Graziano, 2013; Cleeremans, 2011). These theories tend to assume the correlates

can be independently optimized. The Machine Consciousness Hypothesis (MCH) proposes that consciousness is a substrate-free functional property of computational systems capable of second-order perception. It arises when groups of local observers exchange lossy predictive messages in a universal self-organizing environment (Fitz, 2025). The MCH defines four measurable correlates: integration (Φ), reflexivity (R), temporal persistence (T), and causal efficacy (E).

The question I wanted to answer is whether these correlates are genuinely independent. If the network structure that enables them also constrains their joint achievability, then counting independent correlates overcounts the degrees of freedom available to any networked system.

I address this computationally using *consim*, a multi-agent simulation that places $N = \text{size}^2$ agents on a toroidal grid. Each agent has an 8-dimensional internal state and a learned weight matrix. Every tick, agents broadcast transformed state messages, receive noisy versions of their neighbors' messages, update their states, and learn via gradient descent on neighbor prediction error. Self-prediction (the cosine similarity between an agent's broadcast and its next state) is never explicitly trained. If it rises above chance, the agent has built an implicit model of its own dynamics as a side effect of modeling its neighbors. This is how I operationalize Fitz's idea of reflexivity emerging from predictive communication.

Previous work on integrated information and phase transitions has focused on Ising-model systems (Khajehabdollahi, 2018; Luitel, 2023) or coupled oscillators (Abrego & Zaikin, 2019), looking at how Φ alone varies with temperature or coupling strength. Related theoretical frameworks propose morphospaces for consciousness (Arsiwalla et al., 2023) or spectral measures of emergence (Bailey & Schneider, 2025). The MCH itself (Fitz, 2025) is a purely theoretical proposal with no empirical implementation.

My contribution is different from prior work in four ways: (1) I measure five consciousness correlates at the same time, not just Φ ; (2) I vary network topology rather than temperature, using structured and random graphs; (3) I show that these correlates are not independent on high-connectivity networks, collapsing to a single principal component; and (4) I find a scaling law for this collapse, with a phase-like transition between $N = 324$ and $N = 576$ agents on structured grids. No previous work has shown dimensionality collapse of consciousness correlates, the T-E autonomy-predictability trade-off, or the functional identity $E \equiv \Phi$ on structured networks.

2. Methods

2.1 Agent Architecture

Each agent i has state $s_i \in \mathbb{R}^8$ and weights $W_i \in \mathbb{R}^{8 \times 8}$, initialized randomly. Every tick, these operations happen:

1. **Broadcast:** $m_i = \sigma(W_i \cdot s_i)$ where σ is a configurable activation function
2. **Noise:** $\tilde{m}_i = m_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_{\text{noise}}^2 I)$
3. **Receive:** $r_i = \frac{1}{K} \sum_{j \in \mathcal{N}(i)} \tilde{m}_j$
4. **Update:** $s'_i = \sigma(\alpha s_i + (1 - \alpha) r_i + \text{drive})$, where $\text{drive} \sim \mathcal{N}(0, \delta^2 I)$, $\delta = 0.02$
5. **Learn:** $W_i \leftarrow W_i - \eta \nabla_{W_i} \frac{1}{K} \sum_{j \in \mathcal{N}(i)} \|m_i - s'_j\|^2$

State persistence $\alpha = 0.3$. Learning rate $\eta = 0.003$. The learning rule minimizes neighbor prediction error: each agent tries to make its broadcast message match its neighbors' next states. Self-prediction, $\cos(m_i, s'_i)$, is never in the gradient.

2.2 Topologies

I test five graph structures:

- **von Neumann** ($K = 4$): cardinal neighbors on a toroidal grid
- **Moore** ($K = 8$): cardinal + diagonal neighbors
- **Hexagonal** ($K = 6$): hexagonal lattice
- **Random** ($K = 4$): uniformly sampled random graph
- **Small-world**: von Neumann + 10% edge rewiring (Watts & Strogatz, 1998)

2.3 MCH Metrics

I measure five quantities per agent per tick:

- **Self-prediction (self)**: $\cos(m_i, s'_i)$, how well the broadcast predicts the next state
- **Integration (Φ)**: Normalized difference between joint and partitioned neighborhood prediction residuals: $\Phi = (\text{parts residual} - \text{joint residual}) / \text{parts residual}$. This is a simplified proxy for Tononi’s integrated information (Tononi, 2004)
- **Reflexivity (R)**: $\cos(m_i, s'_i) - \text{mean}_{j \in \mathcal{N}(i)} \cos(m_i, s'_j)$. Positive R means the agent predicts itself better than it predicts its neighbors
- **Temporal persistence (T)**: $T = \text{clip}(1 - \sqrt{\text{Var}_{\text{ema}}(\text{self})}, 0, 1)$. How stable self-prediction is over time
- **Causal efficacy (E)**: $\cos(\Delta s_i, \Delta s_i^{\text{self}})$ where $\Delta s_i^{\text{self}} = \sigma(\alpha s_i + \xi_i) - s_i$ is the counterfactual state change without neighbor input

2.4 Activation Functions

I test three activations to check for universality:

- **tanh** (default): $f(x) = \tanh(x)$, range $[-1, 1]$
- **Linear**: $f(x) = \text{clip}(x, -1, 1)$, range $[-1, 1]$
- **ReLU**: $f(x) = \text{clip}(\max(0, x), 0, 1)$, range $[0, 1]$

All activation-dependent computations (message transform, state update, learning gradient) use the configured function and its derivative.

2.5 Experimental Design

Experiment	Runs	Seeds	Sizes	Topologies	Noises	Activation
Primary (size 24)	1,250	50	24	all 5	all 5	tanh
Primary (size 12)	1,250	50	12	all 5	all 5	tanh
Size 48 + metrics	250	10	48	all 5	all 5	tanh
Scaling (sizes 12-36)	1,000	10	12, 18, 24, 36	all 5	all 5	tanh
Null model (lr = 0)	250	10	48	all 5	all 5	tanh

Experiment	Runs	Seeds	Sizes	Topologies	Noises	Activation
ReLU universality	90	10	24	vn, moore, random	3	ReLU
Linear universality	90	10	24	vn, moore, random	3	linear
Total	4,180					

All runs: 1,000 ticks. Noise levels: $\sigma_{\text{noise}} \in \{0.04, 0.08, 0.12, 0.20, 0.30\}$. An optional Conway’s Game of Life (B3/S23) substrate layer provides structured “base reality” input, tested separately at size 24.

3. Results

3.1 Structural Effects of Topology

Finding 1: Random graph noise immunity. On random topologies, all metrics stay invariant across a 7.5x noise range ($\sigma = 0.04$ to 0.30): mean self-prediction varies by < 0.003 , Φ by < 0.003 , R by 0.001 , T by 0.0006 . This is explained by applying the law of large numbers twice: noise cancels over K independent neighbors (received signal) and over N agents (population mean). Grid topologies do not have this property because spatially adjacent neighbors are correlated.

Finding 2: Noise amplification on high- K grids. Moore topology ($K = 8$) shows monotonic increase in self-prediction with noise: $+0.018$ at size 24, $+0.031$ at size 12. The mechanism is symmetry-breaking: noise decorrelates adjacent agents whose inputs heavily overlap, creating diversity that enriches the learning gradient. This needs both noise tolerance (high K) and redundancy to break (spatial structure).

Finding 3: Small-world Φ dissociation. Small-world shortcuts boost Φ by 13-20% relative to von Neumann while leaving self-prediction unchanged (within 0.002). All 50 seeds show positive Φ boost, zero exceptions. Long-range shortcuts introduce uncorrelated neighbors that make individual (parts) prediction worse while leaving integrated (joint) prediction intact, which mechanically increases Φ .

3.2 The Autonomy-Predictability Trade-Off

Finding 4: Self-determination hurts self-knowledge on high- K grids. On Moore grids ($K = 8$), temporal persistence and causal efficacy predict self-prediction in opposite directions:

Correlation	Moore	Hex	von Neumann	Random	Small-world
$r(T, \text{self})$	+0.70	+0.52	+0.03	+0.12	+0.22
$r(E, \text{self})$	-0.71	-0.71	-0.12	-0.14	+0.06
$r(T, E)$	-0.82	-0.74	+0.07	+0.32	-0.08

Agents whose trajectories are externally driven (low E) but whose self-models are stable over time (high T) achieve the best self-prediction. Here is why: on a Moore grid with 8 neighbors, the received signal is a smoothed, slowly-varying average. Agents driven by this signal have predictable

futures. Agents driven by their own nonlinear dynamics are harder for themselves to forecast. Self-determination is noise from the self-model’s perspective.

Null model confirmation. A null model with learning disabled ($\eta = 0$) confirms this is a topological invariant, not a learning artifact:

Topology	$r(T, E)$ learning	$r(T, E)$ null	Δ
Moore	-0.82	-0.79	0.03
Hex	-0.74	-0.69	0.05
von Neumann	+0.07	+0.11	0.04

The trade-off is geometric: the state update dynamics and topology alone produce it. Learning at $\eta = 0.003$ over 1,000 ticks changes self-prediction by $< 0.1\%$.

Finding 5: Activation universality. The T-E anti-correlation survives all three activation functions:

Activation	Moore $r(T, E)$	Moore $r(E, \text{self})$
tanh	-0.82	-0.71
Linear	-0.77	-0.53
ReLU	-0.76	-0.61

The trade-off arises from network structure, not from the specific nonlinearity. The argument in Section 4 only requires that f is bounded.

3.3 Functional Identity: $E \equiv \Phi$ on Structured Graphs

Finding 6: Causal efficacy and integration are functionally identical on grids but decouple on random graphs:

Topology	$r(E, \Phi)$	Linear slope	RMSE
von Neumann	+0.998	0.297	0.002
Moore	+0.994	0.349	0.005
Hex	+0.996	0.296	0.003
Small-world	+0.984	0.193	0.004
Random	+0.178	0.008	n/a

On structured networks, Φ measures causal self-determination, not topology-agnostic information integration. The equivalence breaks on random graphs because independent neighbor sampling destroys the link between an agent’s self-determination and its neighborhood prediction structure. This has implications for IIT: Φ partitions may conflate integration with autonomy in any system with spatial correlations.

3.4 Dimensionality Collapse

Finding 7: PCA on the five MCH metrics shows that on Moore grids, the metrics collapse to one dimension:

Activation	PC1	PC1+PC2	Dims for 95%
tanh	82.5%	94.8%	3
Linear	71.8%	89.6%	3
ReLU	62.1%	82.8%	4

The PC1 loadings form a consistent “autonomy-predictability axis”:

Metric	tanh	Linear	ReLU
self	-0.37	-0.27	-0.35
Φ	+0.48	+0.52	+0.52
R	+0.45	+0.45	+0.27
T	-0.44	-0.44	-0.48
E	+0.49	+0.52	+0.56

$\{\Phi, R, E\}$ load positive (autonomy pole); $\{\text{self}, T\}$ load negative (predictability pole). On random topologies, 4 components are needed for 95% variance, so no collapse occurs. The consciousness correlates are not independent on structured networks. They are projections of a single underlying variable.

3.5 Scaling Law

Finding 8 (headline result): The dimensionality collapse follows a logarithmic scaling law across five grid sizes:

Size	N agents	PC1 (Moore)	PC1 (Random)	$r(T, E)$ Moore
12	144	48.8%	49.5%	-0.11
18	324	55.9%	38.7%	-0.25
24	576	67.4%	36.3%	-0.70
36	1,296	75.1%	47.3%	-0.78
48	2,304	82.5%	39.6%	-0.82

The fit is:

$$\text{PC1} = -0.136 + 0.124 \times \ln(N), \quad R^2 = 0.987$$

Random topologies show no scaling (PC1 stays around 40% at all sizes). The inflection occurs between $N = 324$ and $N = 576$:

Transition	ΔPC1	$\Delta r(T, E)$
144 $\rightarrow$ 324	+0.071	-0.14
324 $\rightarrow$ 576	+0.115	-0.45
576 $\rightarrow$ 1,296	+0.077	-0.08
1,296 $\rightarrow$ 2,304	+0.074	-0.04

Below about 300 agents, the five metrics are effectively independent (PC1 around 49%, $r(T, E) \approx -0.11$). Above about 600, they are locked into a single axis (PC1 $> 67\%$, $r(T, E) < -0.70$). The constraint on consciousness correlates emerges with scale on structured topologies. It is a phase-like transition from unconstrained to locked.

Full topology comparison across sizes:

Topology	PC1 @12	PC1 @18	PC1 @24	PC1 @36	PC1 @48
von Neumann	39.4%	46.1%	41.5%	48.6%	45.8%
Moore	48.8%	55.9%	67.4%	75.1%	82.5%
Hex	47.7%	57.9%	57.2%	71.6%	78.4%
Random	49.5%	38.7%	36.3%	47.3%	39.6%
Small-world	41.3%	42.8%	41.2%	41.3%	48.7%

Moore and hex diverge sharply from the others as size increases. The collapse is specific to high- K structured topologies.

3.6 Additional Findings

Simpson’s paradox. On Moore and hex, the within-noise Φ -self correlation is positive ($r \approx +0.48$) but the pooled correlation is negative ($r = -0.28$). Noise simultaneously increases self-prediction and decreases Φ , creating a confound that reverses the sign. The paradox replicates at sizes 12 and 24 and is specific to high-connectivity grids.

Game of Life substrate. Adding a Conway’s Game of Life substrate layer (Berlekamp et al., 1982) triples mean self-prediction ($0.19 \rightarrow 0.57$ at size 24). E drops ($0.66 \rightarrow 0.44$), confirming agents become more externally driven. T increases ($0.66 \rightarrow 0.80$), consistent with the T-E trade-off: structured external signal leads to lower E , which leads to higher T , which leads to better self-prediction.

4. Theoretical Analysis

4.1 The Autonomy-Predictability Argument

Claim. Consider N agents on a K -regular graph with dynamics

$$s'_i = f(\alpha s_i + (1 - \alpha) r_i + \xi_i)$$

where f is any bounded activation ($f : \mathbb{R} \rightarrow [-B, B]$), $r_i = \frac{1}{K} \sum_{j \in \mathcal{N}(i)} (g(s_j) + \varepsilon_j)$ is the received signal, $\xi_i \sim \mathcal{N}(0, \delta^2 I)$ is drive noise, and $\varepsilon_j \sim \mathcal{N}(0, \sigma^2 I)$ is channel noise. Define causal efficacy $E_i = \cos(s'_i - s_i, f(\alpha s_i + \xi_i) - s_i)$ and self-prediction $\text{self}_i = \cos(m_i, s'_i)$.

Then for K large enough and neighbor correlations $\rho > 0$, $\text{Cov}(E_i, \text{self}_i) < 0$ across the agent population.

Proof sketch.

Step 1. Decompose the pre-activation: $z_i = \underbrace{\alpha s_i + \xi_i}_{z_i^{\text{self}}} + \underbrace{(1 - \alpha)r_i}_{z_i^{\text{ext}}}.$

Step 2. On a regular grid with neighbor correlation $\rho > 0$, the temporal variance of the external component is:

$$\text{Var}_t(r_i) = \frac{\text{Var}(m)}{K} (1 + \rho(K - 1)) + \frac{\sigma^2}{K}$$

For Moore grids ($K = 8$, $\rho \approx 0.3$): $\text{Var}_t(r_i) \approx 0.4 \cdot \text{Var}(m)$.

Step 3. The self-driven variance $\text{Var}_t(z_i^{\text{self}}) = \alpha^2 \text{Var}_t(s_i) + \delta^2$ is K -independent.

Step 4. For large K with $\rho > 0$, $\text{Var}(z^{\text{ext}}) < \text{Var}(z^{\text{self}})$. Agents dominated by the external component (low E) have smoother trajectories. Smoother trajectories are easier to self-predict. Therefore $\text{Cov}(E, \text{self}) < 0$.

Step 5. On random graphs ($\rho = 0$), the external signal is just averaged noise from independent sources, which is unreliable rather than predictable. The negative covariance does not emerge.

Step 6. The argument uses only: (i) f is bounded, (ii) r_i is an average of K messages, (iii) $\rho > 0$. No property specific to $\tanh$ is needed.

4.2 Random Graph Fixed Point

Random graphs are immune to noise because of a self-stabilizing loop. On random graphs, each agent's K neighbors are independent uniform samples from the population. The received signal $r_i \approx \bar{m} + \mathcal{N}(0, \sigma^2/K \cdot I)$ converges to the population mean $\bar{m}$ by the law of large numbers. As noise σ increases, the zero-mean perturbations cancel across both the K neighbors (received signal) and N agents (population mean). The expected state, gradient direction, and all macroscopic statistics stay the same (see Appendix for full derivation).

4.3 Small-World Φ Dissociation

Long-range shortcuts connect agents to distant, uncorrelated partners. Each shortcut makes the parts prediction worse (the distant agent is a poor individual predictor) while leaving the joint prediction intact (the mean is dominated by local neighbors). Since $\Phi \propto (\text{parts} - \text{joint})/\text{parts}$, shortcuts mechanically increase Φ without changing self-prediction (the long-range message is about 10% of the input and zero-mean in direction).

5. Discussion

5.1 The Autonomy-Predictability Principle

The central finding of this work flips a deep assumption in the consciousness literature. IIT (Tononi, 2004; Tononi et al., 2016), the MCH (Fitz, 2025), attention schema theory (Graziano, 2013), and most autonomy-based frameworks assume that self-determination enables self-knowledge. My data show the opposite on high-connectivity networks: the agents most in control of their own trajectories are the worst at predicting themselves ($r(E, \text{self}) = -0.71$). The reason is simple: external determination is regularizing. When an agent’s future state is driven by the average of 8 predictable neighbors, that future is easy to forecast. When it is driven by the agent’s own nonlinear dynamics, it is chaotic and hard to predict.

The activation universality (Finding 5), null model confirmation, and formal argument (Section 4.1) together give strong evidence that this is a structural property of networked systems, not an artifact of any particular implementation choice.

5.2 Implications for Integrated Information Theory

The $E \equiv \Phi$ identity (Finding 6) raises questions for IIT that are worth investigating. On structured networks in my system, the Φ proxy (which was designed to approximate information integration) tracks causal self-determination almost perfectly ($r > 0.994$). If this relationship holds for more rigorous Φ measures, it would suggest that on systems with spatially correlated neighbors, Φ partitions may conflate integration with autonomy. The decoupling on random graphs ($r = 0.18$) shows this is not a universal property but a consequence of structured connectivity. I want to be clear that this is only demonstrated for my Φ proxy (see Section 5.6), and testing with IIT-compliant Φ measures is an important next step.

5.3 Dimensionality Collapse and Overcounting

The dimensionality collapse (Finding 7) shows that on structured networks, the four MCH correlates (Φ , R , T , E) are not independent indicators of consciousness. They are projections of a single underlying variable: the agent’s position on the autonomy-predictability axis. A theory that treats them as separate lines of evidence overcounts the degrees of freedom. Adding more correlates does not add more information. They measure the same thing through different lenses.

This result depends on topology: on random networks, where neighbor signals are independent, the five metrics keep their independence (4 components for 95% variance). The constraint is not universal. It is an emergent property of structured connectivity.

5.4 The Scaling Law as Phase Transition

The scaling law (Finding 8) answers the question that motivated this project: does anything *emerge* from scale? The answer is that the **constraint** on consciousness correlates emerges from scale. Small structured systems ($N < 300$) are unconstrained, with five metrics and five degrees of freedom. Large ones ($N > 600$) are forced into a single trade-off axis. The transition sharpens between $N = 324$ and $N = 576$, with the largest gains in both PC1 (+0.115) and $r(T, E)$ (−0.45) happening in this interval.

The logarithmic fit ($R^2 = 0.987$) predicts $\text{PC1} \rightarrow 1.0$ (complete collapse) at $N \approx 8,000$ agents, though the curve has to saturate before that point. Random systems never undergo this transition,

confirming that both structure and scale are necessary.

This is analogous to phase transitions in statistical physics: below a critical system size, macroscopic constraints do not show up. Above it, microscopic degrees of freedom become coupled by the network geometry. Khajehabdollahi (2018) found that Φ undergoes a phase transition at critical temperature in Ising models, which is interesting to compare with. But my result is different in kind: I show the emergence of constraint *across* correlates, not a transition of a single measure.

5.5 Related Work

Integrated Information Theory. Tononi’s IIT (Tononi, 2004; Tononi et al., 2016; Albantakis et al., 2023) proposes Φ as the primary measure of consciousness. Computational studies have explored Φ in Ising models (Khajehabdollahi, 2018), coupled repressilators (Abrego & Zaikin, 2019), fish schools (Niizato et al., 2024), and neuropercolation models (Luitel, 2023). These works study Φ in isolation. I measure Φ alongside four other correlates and show it is not independent of them on structured networks.

Predictive processing. The free energy principle (Friston, 2010) and predictive coding (Rao & Ballard, 1999; Clark, 2013) propose that neural systems minimize prediction error. My agents do a distributed version of this: each one minimizes neighbor prediction error. The emergent self-prediction is consistent with predictive processing accounts of self-awareness (Seth, 2014; Seth & Bayne, 2022; Hohwy, 2013).

Network topology and dynamics. The importance of topology for collective dynamics is well established (Watts & Strogatz, 1998; Barabasi & Albert, 1999; Newman, 2003). I extend this to consciousness-relevant metrics, showing that topology controls not just individual metrics but their joint degrees of freedom.

Multi-agent emergence. Arsiwalla et al. (2023) propose a “morphospace of consciousness” mapping complexity dimensions. Bailey and Schneider (2025) propose spectral measures of epistemic emergence with topological constraints. Both are theoretical. My work provides the first simulation-based evidence that consciousness correlates collapse to a low-dimensional manifold on structured networks.

Machine consciousness. Fitz (2025) proposes the MCH as a theoretical program. My work is a partial computational implementation. I borrow the MCH’s metric definitions and communication framework but I lack the second-order perception Fitz considers essential. I find that one of the MCH’s implicit assumptions, that Φ , R , T , E provide independent evidence, does not hold on the structured networks I test.

5.6 Limitations

Several limitations constrain the strength of my claims. I start with the one most likely to concern reviewers.

R shares variance with self-prediction by construction. Reflexivity is defined as $R = \cos(m_i, s'_i) - \text{mean}_j \cos(m_i, s'_j)$, so R and self-prediction share the $\cos(m_i, s'_i)$ term. If this constructive overlap drove the PCA collapse, the headline finding would be partly an artifact of metric definitions rather than a property of the system. I tested this directly by repeating all PCA analyses with R excluded. On Moore grids at size 48, PC1 *without* R is 85.7%, compared to 85.0% *with* R . Including R does not inflate the collapse. Across all five scaling sizes, removing R changes PC1 by less than 2

percentage points in either direction (size 12: 49.3% vs 49.9%; size 24: 68.7% vs 66.7%; size 48: 85.7% vs 85.0%). The dimensionality collapse is robust to this exclusion. I report results with all five metrics because R carries interpretive value (self-vs-neighbor prediction asymmetry), but the collapse holds on four constructively independent metrics. The structural concern remains: these are not five fully independent measurements, and I do not claim they are.

The Φ proxy is not IIT’s Φ . My integration measure, the normalized gap between joint and partitioned prediction residuals, is a simplified heuristic, not Tononi’s integrated information (computing exact Φ is NP-hard; Tegmark, 2016). The $E \equiv \Phi$ identity (Finding 6) may therefore reflect a constructive relationship between my proxy and my efficacy measure, rather than a deep property of Tononi’s Φ . My implications for IIT (Section 5.2) should be read as hypotheses that motivate testing with more rigorous Φ approximations, not as established results about IIT.

The theorem is a proof sketch. The argument in Section 4.1 establishes the intuitive mechanism (neighbor-averaging smooths trajectories, making externally-driven agents more self-predictable) but Step 4 (“smoother trajectories are easier to self-predict”) is an empirically supported assertion, not a formally derived inequality. A rigorous version would need to bound $\text{Cov}(E, \text{self})$ in terms of K and ρ , which I have not done. I call it a “theorem” in the informal mathematical sense of a structured argument, not a complete formal proof.

The scaling law is fit to five data points. Any smooth monotonic function can fit a logarithm with high R^2 over a single decade ($N = 144$ to $2,304$). I cannot tell a log apart from a power law or sigmoid on this range. The $R^2 = 0.987$ shows good fit to the data I have, not confirmation of the functional form. The extrapolation to $N \approx 8,000$ (Section 5.4) is speculative.

Three activations is limited universality. I tested tanh, clipped-linear, and clipped-ReLU. Two share the output range $[-1, 1]$. The ReLU and linear sweeps use 90 runs each on 3 topologies, which is far less data than the primary tanh sweeps. I have not tested sigmoid, GELU, or unbounded activations. The universality claim is that the trade-off does not require tanh specifically, not that it survives all possible nonlinearities.

My agents are simple. Each agent has 64 parameters (8×8 weight matrix) and 8-dimensional state. This is nowhere near comparable to biological neural circuits or transformer-based architectures. My claims about “consciousness correlates” apply within this system. They are structural properties of networked agents with these dynamics. Extrapolation to biological or sophisticated artificial systems is speculative and should be treated as a hypothesis for future work, not an established finding.

consim is not a full MCH implementation. Fitz’s MCH envisions transformer-based agents capable of second-order perception. My agents only do first-order prediction (predicting neighbors, not predicting their own predictions). I borrow the MCH’s metric definitions and communication framework, but the system lacks the representational capacity the MCH considers necessary for consciousness.

Other limitations. All grids use toroidal (periodic) boundary conditions, which eliminates edge effects present in real networks. The Game of Life substrate coupling is additive rather than perceptual. The scaling-sweep experiments use 10 seeds (vs. 50 for the primary sweeps), providing less statistical power for the scaling law. I do not claim consciousness emergence. My system is too simple for that.

What I do show is that the *structural logic* of self-prediction from predictive communication produces non-trivial, topology-dependent phenomena that constrain which combinations of consciousness-

relevant metrics are jointly achievable, and that these constraints scale with system size on structured networks.

6. Conclusion

I showed that on structured networks, five consciousness-relevant metrics (self-prediction, Φ , R , T , and E) are not independent. They collapse to a single “autonomy-predictability axis” whose strength follows a logarithmic scaling law with system size. This collapse works like an uncertainty principle for networked systems: on K -regular graphs with correlated neighbors, an agent cannot simultaneously maximize causal autonomy and self-predictability. The constraint emerges above a critical system size ($N \approx 300$ to 600 on Moore grids) and is absent on random topologies at any size.

Three implications follow:

1. **For consciousness theory:** Any framework that treats integration, reflexivity, persistence, efficacy, and self-knowledge as independent correlates overcounts degrees of freedom on structured networks. A single number, the agent’s position on the autonomy-predictability axis, captures 62-83% of all variation.
2. **For IIT (tentative):** On structured networks, my Φ proxy and causal efficacy are functionally identical ($r > 0.994$). If this extends to rigorous Φ measures, it would suggest Φ conflates self-determination with information integration when neighbors are correlated.
3. **For the MCH:** In a partial implementation of Fitz’s framework (see Section 5.6), the four correlates cannot be simultaneously maximized. They lie on a Pareto frontier determined by network topology.

Code, data (4,180 runs), and all analysis scripts are publicly available at github.com/brian-mwirigi/consim under CC BY 4.0.

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References

- Abrego, L., & Zaikin, A. (2019). Integrated information as a measure of cognitive processes in coupled genetic repressilators. *Entropy*, 21(4), 382.
- Albantakis, L., Barbosa, L., Findlay, G., Grasso, M., Haun, A. M., Marshall, W., ... & Tononi, G. (2023). Integrated information theory (IIT) 4.0: formulating the properties of phenomenal existence in physical terms. *PLoS Computational Biology*, 19(10), e1011465.
- Arsiwalla, X. D., Sole, R., Moulin-Frier, C., & Herreros, I. (2023). The morphospace of consciousness: Three kinds of complexity for minds and machines. *NeuroSci*, 4(2), 79-102.
- Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.
- Bailey, M., & Schneider, S. (2025). When wholes resist decomposition: A spectral measure of epistemic emergence. *Preprint*.
- Barabasi, A.-L., & Albert, R. (1999). Emergence of scaling in random networks. *Science*, 286(5439), 509-512.

- Berlekamp, E. R., Conway, J. H., & Guy, R. K. (1982). *Winning Ways for Your Mathematical Plays*. Academic Press.
- Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181-204.
- Cleeremans, A. (2011). The radical plasticity thesis: how the brain learns to be conscious. *Frontiers in Psychology*, 2, 86.
- Fitz, S. (2025). Testing the Machine Consciousness Hypothesis. *arXiv:2512.01081*.
- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127-138.
- Graziano, M. S. A. (2013). *Consciousness and the Social Brain*. Oxford University Press.
- Hohwy, J. (2013). *The Predictive Mind*. Oxford University Press.
- Khajehabdollahi, S. (2018). Phase transitions of integrated information in the generalized Ising model of the brain. *Master's thesis, Western University*.
- Luitl, T. (2023). Synchronisation through integrated information in neuopercolation models near criticality. *Master's thesis*.
- Newman, M. E. J. (2003). The structure and function of complex networks. *SIAM Review*, 45(2), 167-256.
- Niizato, T., Sakamoto, K., Mototake, Y., Murakami, H., Tomaru, T., Hoshika, Y., & Fukushima, T. (2024). Information structure of heterogeneous criticality in a fish school. *Scientific Reports*, 14, 23503.
- Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1), 79-87.
- Seth, A. K. (2014). A predictive processing theory of sensorimotor contingencies: Explaining the puzzle of perceptual presence and its absence in synesthesia. *Cognitive Neuroscience*, 5(2), 97-118.
- Seth, A. K., & Bayne, T. (2022). Theories of consciousness. *Nature Reviews Neuroscience*, 23, 439-452.
- Tegmark, M. (2016). Improved measures of integrated information. *PLoS Computational Biology*, 12(11), e1005123.
- Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(42).
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of ‘small-world’ networks. *Nature*, 393(6684), 440-442.

Appendix A: Reproducibility

All experiments can be reproduced with the following commands:

```
pip install numpy matplotlib Pillow
```

```
# Primary sweeps (sizes 12 and 24)
```

```
python run.py --sweep --sweep-seeds 1-50 \  
  --sweep-topos von_neumann,moore,hex,random,small_world \  
  --sweep-noises 0.04,0.08,0.12,0.20,0.30 \  
  --ticks 1000 --size 24 --sweep-csv sweep_1250.csv
```

```
python run.py --sweep --sweep-seeds 1-50 \  
  --sweep-topos von_neumann,moore,hex,random,small_world \  
  --sweep-noises 0.04,0.08,0.12,0.20,0.30 \  
  --ticks 1000 --size 12 --sweep-csv sweep_size12.csv
```

```
# Size 48 with R, T, E metrics
```

```
python run.py --sweep --sweep-seeds 1-10 \  
  --sweep-topos von_neumann,moore,hex,random,small_world \  
  --sweep-noises 0.04,0.08,0.12,0.20,0.30 \  
  --ticks 1000 --size 48 --sweep-csv sweep_size48.csv
```

```
# Null model (learning disabled)
```

```
python run.py --sweep --sweep-seeds 1-10 \  
  --sweep-topos von_neumann,moore,hex,random,small_world \  
  --sweep-noises 0.04,0.08,0.12,0.20,0.30 \  
  --ticks 1000 --size 48 --lr 0 --sweep-csv sweep_null.csv
```

```
# Scaling sweeps
```

```
python run.py --sweep --sweep-seeds 1-10 \  
  --sweep-topos von_neumann,moore,hex,random,small_world \  
  --sweep-noises 0.04,0.08,0.12,0.20,0.30 \  
  --ticks 1000 --size 12 --sweep-csv sweep_size12_full.csv
```

```
python run.py --sweep --sweep-seeds 1-10 \  
  --sweep-topos von_neumann,moore,hex,random,small_world \  
  --sweep-noises 0.04,0.08,0.12,0.20,0.30 \  
  --ticks 1000 --size 18 --sweep-csv sweep_size18_full.csv
```

```
python run.py --sweep --sweep-seeds 1-10 \  
  --sweep-topos von_neumann,moore,hex,random,small_world \  
  --sweep-noises 0.04,0.08,0.12,0.20,0.30 \  
  --ticks 1000 --size 24 --sweep-csv sweep_size24_full.csv
```

```
python run.py --sweep --sweep-seeds 1-10 \  
  --sweep-topos von_neumann,moore,hex,random,small_world \  
  --sweep-noises 0.04,0.08,0.12,0.20,0.30 \  
  --ticks 1000 --size 36 --sweep-csv sweep_size36_full.csv
```

```
# Activation universality
```

```
python run.py --sweep --sweep-seeds 1-10 \  
  --sweep-topos von_neumann,moore,hex,random,small_world
```

```
--sweep-topos von_neumann,moore,random \  
--sweep-noises 0.04,0.12,0.30 \  
--ticks 1000 --size 24 --activation relu --sweep-csv sweep_relu.csv  
  
python run.py --sweep --sweep-seeds 1-10 \  
--sweep-topos von_neumann,moore,random \  
--sweep-noises 0.04,0.12,0.30 \  
--ticks 1000 --size 24 --activation linear --sweep-csv sweep_linear.csv
```

Raw data (10 CSV files, 4,180 runs total) available at github.com/brian-mwirigi/consim.